THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A zigzag graph with varying slopes shows: 1

(a) rest **(b)** non-uniform motion
(c) uniform motion **(d)** periodic only

2. Which of the following motion is uniform? 1

(a) a ball in a game **(b)** the second hand of a clock
(c) a bus in a city **(d)** a kite in a gust

3. A man walks 3 km at 6 km/h and then 3 km at 3 km/h. The total time is: 1

(a) 3 h **(b)** 2 h **(c)** 1.5 h **(d)** 1 h

4. Time period of a pendulum is measured more accurately by: 1

(a) guessing **(b)** measuring half oscillation
(c) measuring 20 oscillations **(d)** measuring 1 oscillation

5. A body moving with uniform speed covers 20 m in 4 s. In 10 s it will cover: 1

(a) 50 m **(b)** 40 m **(c)** 80 m **(d)** 20 m

6. Which of the following is the greatest speed? 1

(a) 20 m/s **(b)** 15 m/s **(c)** 50 km/h **(d)** 90 km/h

7. When the length of a pendulum is increased, its time period: 1

(a) increases **(b)** becomes 0 **(c)** stays same **(d)** decreases

8. A boy takes 30 minutes to walk to school at 4 km/h. The distance is: 1

(a) 2 km **(b)** 120 km **(c)** 0.5 km **(d)** 8 km

9. Which speed is the smallest? 1

(a) 3 m/s **(b)** 20 m/s **(c)** 36 km/h **(d)** 18 km/h

10. The bob's mass of a pendulum is doubled. The time period: 1

(a) becomes zero **(b)** remains same **(c)** halves **(d)** doubles

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A body moves 20 m in 4 s and then rests for 6 s. Find average speed for 10 s. 2

12. Find the speed of a runner who covers 2 km in 8 min in km/h. 2

13. Explain how the odometer reading is used to find distance. 2

14. Why are digital clocks more precise than sundials? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how to find the time period of a pendulum accurately and show with 20 oscillations in 36 s. 3

16. A student times 10 oscillations of a pendulum as 16 s and 20 oscillations as 32 s. Show that the time period is the same and find it. 3

17. Explain with an example how time period of a pendulum can be used as a time unit. 3

OR

The distance-time data: 0 s 0 m; 2 s 8 m; 4 s 16 m; 6 s 24 m; find speed and state whether it is uniform. 3

18. Why is the motion of a swing periodic but not uniform? 3

OR

Why is the sundial inaccurate on a cloudy day? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Distinguish between uniform and non-uniform motion using the example of a train starting from a station, running on a straight track and stopping. 5

OR

Explain average speed using a round trip where outward speed is 30 km/h and return speed is 60 km/h over 60 km each way. 5

20. A student walks 900 m in 10 min, then runs 900 m in 3 min. Find the speeds in m/s and average speed. 5

OR

Describe the use of the distance-time graph in analysing a cyclist's ride from the data given in a table. 5

END OF QUESTION PAPER Total: 40 Marks

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